



FINAL YEAR PROJECT MANUAL

Version 1.0



Institute of Computing
Kohat University of Science and Technology, Kohat-26000
Khyber Pakhtunkhwa, Pakistan

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1. FYP Panel

There are five faculty members in FYP Panel. For any type of evaluation at least three members are to be present. Any evaluation by less than three members will not be accepted.

S.No	Name	Role
1.	Dr. Muhammad Irfan Uddin	Coordinator
2.	Dr. Shafaq Mussadiq	Member
3.	Dr. Muneer Umar	Member
4.	Mr. Muhammad Roman	Member
5.	Mr. Muhammad Ali Zeb	Member

2. FYP re-evaluation committee

S.No	Name	Role
1.	Dr. Shafiullah	Convenor
2.	Dr. Asad Habib	Member
3.	Dr. Saima Hassan	Member
4.	Dr. Abdul Shahid	Member

3. FYP Calendar

Following is the calendar for FYP and will be strictly followed throughout the project.

Time	Activity
Proposal	
Last week of fifth semester	Faculty/Students submit projects' ideas to Panel Group Selection and Supervisor Selection
Second week of sixth semester	Projects Ideas evaluation by Panel
Third week of sixth semester	Final Proposal Selection
Fourth week of sixth semester	FYP Seminar
Fifth week of sixth semester	Proposal Preparation
Three weeks before seventh semester	Ring Binding of Proposal and Turnitin report
Second week of seventh semester	Proposal Defence
Sixth week of seventh semester	Proposal Re-defence (For major revisions/not approved)
Evaluation I	
Third Week: Seminar 1	
Third week of seventh semester	Background study
Fourth week of seventh semester	Background study
Fifth week of seventh semester	Prepare mock-up of full system
Sixth week of seventh semester	Prepare DFD until more refined levels
Sixth Week: Seminar 2	
Seventh week of seventh semester	Prepare Class Diagram
Seventh week of seventh semester	Put all items above in a document form
Eighth week of seventh semester	Submit ring binding of Chapters 1 to 3 along with student and supervisor evaluation forms and Turnitin report
Deliverables:	During ninth week of seventh semester Panel will meet to evaluate the submitted materials. Marks out of 50 are assigned in KCMS. No need for student presentation.
- Chapter 1 – Introduction - Chapter 2 – Background Study	



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Chapter 3 – Requirements and proposed system	
Mid Defense/Evaluation II	
Ninth Week: Seminar 3	
Ninth week of seventh semester	Prepare use-case and activity diagrams
Tenth week of seventh semester	Prepare sequence diagrams
Twelve Week: Seminar 4	
Thirteenth week of seventh semester	ERD
Fifteenth week of seventh semester	Prepare Front End of the system with GUI design
Fifteenth week of seventh semester	Explain every item above in a document form
Sixteenth week of seventh semester	Submit ring binding of Chapter 4 along with student and supervisor evaluation forms and Turnitin report
Next day after Final Exam	Mid Defence
Deliverables: Chapter 4 – Analysis and Design	Marks out of 100 are submitted in KCMS
Evaluation III	
Third Week: Seminar 5	
Third week of eighth semester	Study and practice design patterns and best practices
Fourth week of eighth semester	Understand S.O.L.I.D principles
Fifth week of eighth semester	Coding
Sixth Week: Seminar 6	
Sixth week of eighth semester	Coding
Seventh week of eighth semester	Coding
Seventh week of eighth semester	Write your implementation in text
Eighth week of eighth semester	Submit Ring binding of Chapter 5 along with student and supervisor evaluation forms and Turnitin report
Deliverables: Chapter 5 – Implementation	During ninth week of eighth semester, Panel will meet to evaluate the submitted materials. Marks out of 50 are assigned in KCMS. No need for student presentation.
Final Evaluation/Evaluation IV	
Ninth Week: Seminar 7	
Ninth week of eighth semester	Testing and Integration
Twelve Week: Seminar 8	
Tenth week of eighth semester	Testing and Integration
Thirteenth week of eighth semester	Project Showcase (10 Marks)
Fifteenth week of eighth semester	End-to-end testing
Fifteenth week of eighth semester	End-to-end testing
Sixteenth week of eighth semester	Submit Ring binding of Chapters 6 – 7 along with student and supervisor evaluation forms and Turnitin report
Actual: June 12, 2020	
Next day after Final Exam	Final Defence
Deliverables: - Chapter 6 – Testing and Evaluation - Chapter 7 – Conclusion and Future Work	Marks out of 100 are submitted in KCMS

4. FYP Terms of References

In the whole process of FYP, Terms of References (TORs) given below are followed.



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1. The current FYP panel is created on **August 19, 2019** and is valid for next two years until **August 31, 2021**, when it will be automatically dissolved and Director IoC will create a new FYP panel along with new re-evaluation committee.
2. All documents related to FYP are available online for everybody at location <http://tiny.cc/iocfyp>
3. FYP calendar will be strictly followed for all deadlines of FYP.
4. Rules and regulations explained in Manual will be followed as it is in Fall and Spring Semesters for CS491 and CS492 respectively. New changes will be possible after one year, when next fall semester is about to begin.
5. Coordinator should have the right to replace a panel member with another faculty member when there is lack of cooperation from a panel member.
6. Director IoC is requested to assign maximum two courses to FYP panel members, so they are able to follow the process systematically and deadlines are achieved within the schedule.
7. Evaluation
 - a. Three out of five members must be present during any evaluation. Any evaluation by less than three member will be automatically invalid. This is based on the guidelines of Academic Council.
 - b. In case there are less than three panel members, then evaluation will stop. Director IoC must assign a faculty member to complete the panel and hence will be responsible for any delay in result submission.
 - c. In case a panel member is a supervisor, he/she should not participate in evaluation. The rest of the members will evaluate the project. This is based on the guidelines of Academic Council.
 - d. Coordinator can categorize projects according to the nature of work (e.g. web app, mobile app, AI project, etc.) and can assign a panel member as Expert based on the expertise of the member. In case there is a project that is out of the scope of all panel members, then coordinator must request another faculty member (may be external) as Expert.
8. Supervisor load and responsibility
 - a. Students must be equally divided amongst the faculty. During proposal defence, Panel member will ensure that none of the faculty member is assigned more than four projects. In case a supervisor has more than four projects, then Coordinator will inform Director to assign the project to different supervisor.
 - b. Supervisor will have at least bi-weekly meetings with students to guide them and ensure that they are going in a right direction. This progress will be checked in the supervisor evaluation form.
 - c. Supervisor must ensure to provide at least one hour per week or two hours per two weeks to every project under his/her supervision.
 - d. During every evaluation the group of students will present student and supervisor evaluation forms. Supervisor gives comments about the progress of students. Similarly students also give feedback about the supervision of supervisor and inform the panel about their progress. Panel will discuss both of the reports during evaluation.
 - e. In case of unsatisfactory supervision reported by students, Coordinator will report to Director to look into the matter further. Director may change supervisor or ensure that the current supervisor is giving enough time to students.
 - f. Thesis plagiarism must be checked by supervisor in Turnitin and a report is attached with every submission. Plagiarism of maximum 20% is acceptable.
 - g. Supervisor does not assign any marks to own students, as per guidelines of Academic Council.
9. Delayed submission or delayed evaluation



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- a. Delayed submission may be accepted with acceptable excuses approved from Director and with a fine of 100 Rs per day per student. However, no delayed submission will be entertained when defence is actually started.
- b. Student proposal defence schedule must be notified two days before the actual defence. Students must be present in their allotted time. In case students come later than the allotted time, they must request Director with acceptable excuses and a fine of 500 Rs per day per student is imposed counting from the first day of defence until the last day of defence. The panel will reschedule their defence on the last day of evaluation. However, students not showing up until the last day of evaluation, will get Incomplete grade and will re-register the course in next seventh semester.
- c. Student mid defence schedule must be notified two days before the actual defence. Students must be present in their allotted time. In case students come later than the allotted time, they must request Director with acceptable excuses and a fine of 500 Rs per day per student is imposed. The panel will reschedule their mid-defence on the last day of evaluation. However, students not showing up until result is submitted in KCMS, then the Incomplete grade is given. They will appear in the final defence where they will also complete their mid defence.
- d. Students final defence schedule must be notified two days before the actual defence. Students must be present in their allotted time. In case students come later than the allotted time, they must request Director with acceptable excuses and a fine of 500 Rs per day per student is imposed. Their defence, will be re-scheduled to the last day of evaluation. In case a student does not show up until the submission of result in KCMS, then incomplete grade is given and will be re-evaluated in the seventh semester with next FYP mid defence.
- e. Fine is submitted in Director IIT account number 0003437100001103 along with Late Submission/Defence Form.

10. Failed students cases

- a. In case a student fails proposal defence the first time, a second proposal defence will be scheduled within a month. Students failing for second time in proposal defence will be awarded incomplete grade and will have to re-register the course again. Incomplete grade is assigned based on the guidelines of Academic Council.
- b. In case a student fails the mid defence, an Incomplete grade is awarded and re-defence will be at the end of the next semester along with final defence of the same batch. Incomplete grade is given because of Academic Council guidelines.
- c. In case a student fails the final defence, an Incomplete grade is awarded and re-defence will be at the end of the next semester along with mid defence of next batch. Incomplete grade is given because of Academic Council guidelines.

11. Students objection

- a. In case students are not satisfied about the supervision, they have the option to change supervisors. They must submit a written report to Director on the Change Supervisor Form. Director can then decide about changing the supervisor. However, this change is allowed only once in FYP duration.
- b. Students have the right to object any panel member. However, a student can object on maximum two panel members. In this situation the other members will evaluate the project. Student objection must reach to coordinator on the Panel Member Objection Form approved from Director two weeks before the evaluation. Any objection later than that will not be entertained.
- c. Students have the right to object their evaluation. In this case, student will submit a written request to Director and a fee of 500 Rs per student on the Re-evaluation Form. Director IoC will forward the



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case to re-evaluation committee. The re-evaluation committee, will further investigate the case based on the panel evaluations and progress reports. The committee has the right to arrange a re-defence and update the marks in KCMS recorded in [Re-evaluation Decision Form](#). None of the FYP panel member should be in re-evaluation committee.

12. Once students entered their data and decide about a team at the time of proposal submission, then the group will stay as it is until the end of the project. In case there is misunderstanding between group members and they want to change the group, then they must submit a written request to FYP Panel on the [Group Change Form](#). The panel will then decide the case. However, no change is possible after the Evaluation I. In case students want to change the project by any cost, then they must register CS491 and start from a new proposal defence. Group change is allowed only once in FYP session.
13. As per academic council guidelines, students who are registered in CS491 are requested to take maximum of four other courses and when registered in CS492, should take maximum three other courses, to be able to focus on their project and perform better during evaluation. Director will ensure this at the time of registration of courses.
14. During the process of FYP, Coordinator can take any decision subject to rules and regulations of KUST.

5. Proposals

Pool of projects' Ideas

In the last week of the fifth semester, all students registered in fifth semester and all faculty members will be asked to submit at least two projects' ideas, in the format given in [Project Idea Submission Form](#) (c.f. appendixes). If faculty has already discussed their ideas with students, then students' names are also mentioned. If students names are not mentioned then any student can pick the projects' ideas or the panel can allocate the project to any student in case students have difficulty in deciding about a project. Students will mention supervisor name, if they have already discussed, or leave it blank if they want the panel to decide their supervisor. Students are encouraged to submit two projects' ideas, the second idea is as plan B in case first idea is not practical.

During the project idea submission form, students must decide the group they are going to work with. A minimum of two students and a maximum of three students can form a group. We do not allow a single person to take a project, because we encourage students to work in team and learn all the skills of project management in a team. Students' projects are evenly divided by all faculty members i.e. maximum four groups are allowed per faculty member. In case students cannot find supervisor, then panel will assign supervisor.

Projects' Ideas evaluation by Panel

In the second week of the sixth semester, panel evaluates all submitted projects and decides which projects' ideas are to be recommended to be carried out. The panel also records their comments if a particular project idea can be further improved. By submitting the idea to the pool all faculty and students are declaring no conflict of interest and that their knowledge/idea can be shared with other students in FYP.

FYP Seminar

In the fourth week of the sixth semester a seminar is arranged to explain students the process and guidelines of FYP. The schedule of the seminar will be notified in the first week of the sixth semester. All faculty members are also advised to attend the seminar. By the time seminar is arranged, all students must have taken their projects' ideas and have selected their supervisors. A template PowerPoint slide ([Seminars/FYP Seminar.pptx](#)) is prepared for coordinator to present to students.



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Proposal Preparation

From the fifth week of the sixth semester, students start working on proposal. Student will write proposal only on the projects' ideas given in the pool. They are given two months to prepare the proposal.

Proposal Submission

Three weeks before the start of the seventh semester, all proposals must be submitted to the panel. The proposal must be signed by the supervisor. Plagiarism report by Turnitin must be submitted along with proposal. Proposal can only be submitted in the template document prepared by the panel and should not be more than five pages. A PowerPoint template is also prepared to let student know what they have to tell the panel about the project. It is advised that the number of slides is not be more than 15 i.e. one minute on average per slide. Documents related to proposal can be found in [Proposal directory](#).

Proposal Evaluation

At the second week of the seventh semester all submitted proposals are evaluated as per the parameters given in [Proposal Evaluation Form](#) (c.f. appendixes). At least three panel members will be present for evaluation of proposal marking out of 100. The marks are then averaged by the coordinator and decision is recorded in the format given in [Proposal Decision Form](#) (c.f. appendixes).

The presentation should not be more than 30 minutes and we should strictly follow the time. 20 minutes for student presentation and 10 minutes for Q&A. No question is asked by anyone during the presentation, all questions are asked at the end. All students in the group must be ready for the presentation. The panel can randomly ask any group member for presentation.

The decision of the panel about proposal evaluation is binary i.e. proposal is approved or not approved. Proposals approved are two types, i.e. approved as it is and approved with minor revision. In case of minor revision all comments are checked by the supervisor and re-submitted to the panel. No further presentation is required. For approved as it is, at least 70% average marks are obtained and for minor revision at least 60% average marks are obtained. In case of not approved we have two cases i.e. major revisions and not approved at all. In case of major revisions, the group must appear for presentation before the panel and explain how the comments raised by the panel are addressed. For major revisions at least 50% average marks are obtained on average. In case of not approved at all, the group have to work on a new project idea. For major revisions and not approved cases there will be a second defence. Any major revision or not approved for second time means that students will have to re-register the semester. Once the proposal is approved by the panel, only then students can start working on the project.

Proposal Re-evaluation

At the sixth week of seventh semester, panel will re-evaluate the proposals with major revisions and not approved. Any student getting major revisions or not approved in second evaluation will have to re-register the course CS491 in the next semester.

Proposal Flowchart

The process from selecting a project's idea to proposal defence is shown in flow chart given in Figure 1.



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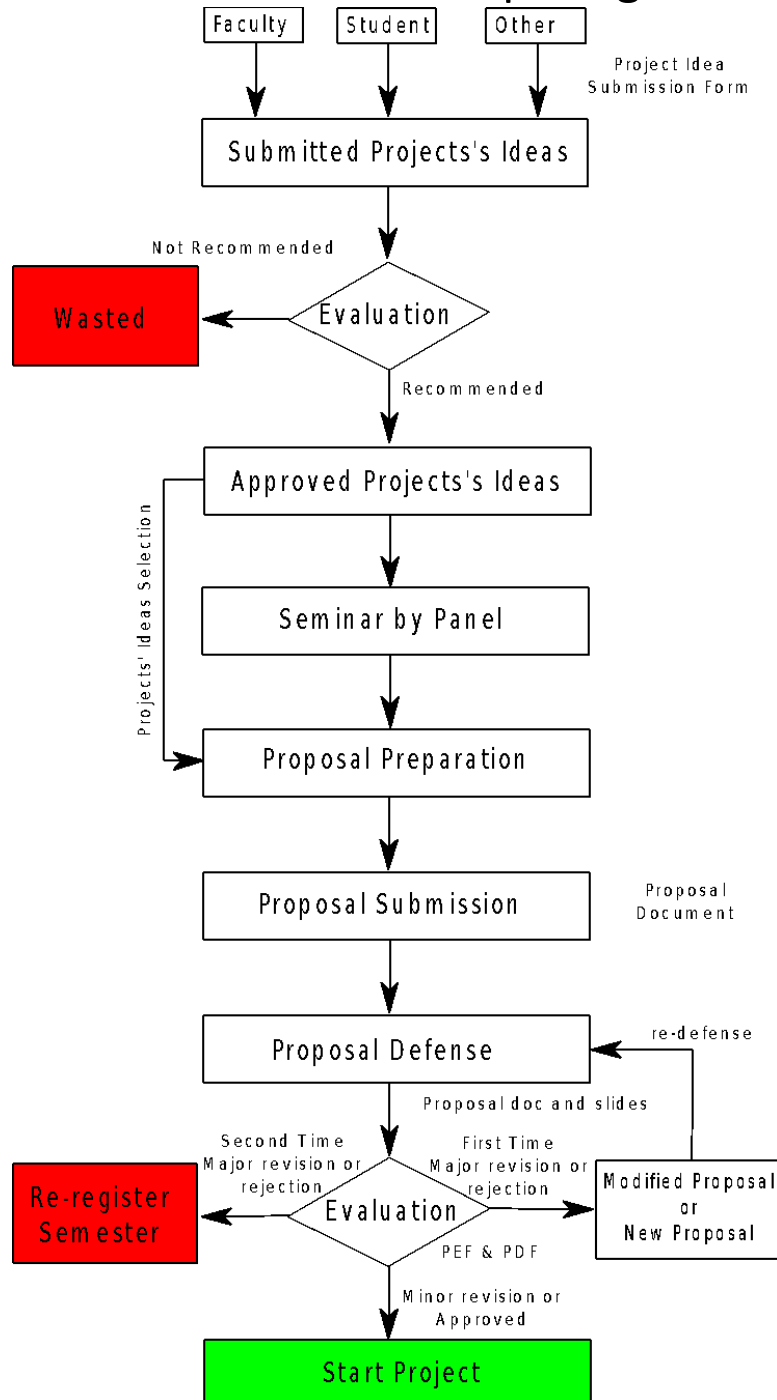


Figure 1: The flow chart showing the whole process from project's idea to proposal defence

Copyright

All projects in the pool are visible to only students registered in FYP and faculty. Neither student nor faculty is allowed to share the information to non-relevant individuals. All students registered in FYP and their supervisors by submitting their proposals declare that there is no conflict of interest amongst them and that their knowledge/ideas can be shared with other students in FYP. All final year projects are intellectual property of Institute of Computing, KUST and should not be used for commercial purposes by either students or faculty.

6. Evaluation Criteria of Project

As per KUST semester rules, the FYP Panel shall use the following percentage for evaluating a project:



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- Background studies (20%)
- Planning and initiation (20%)
- Project work (40%)
- Results and presentation (20%)

There are 6CHs for FYP and hence 200 marks are assigned. The distribution of marks as per the guidelines of Academic Council is given below.

Background studies (20%)

- Software Development Life Cycle (SDLC) (10)
- Software Development Model (10)
- Mock-up (10)
- Documentation 1 (10)

Total Marks = 40

Percentage = $40/200 \times 100 = 20\%$

Planning and initiation (20%)

- Data Flow Diagrams and Class Diagrams (10)
- Use case diagram and activity diagrams (10)
- Sequence diagrams (10)
- Entity Relationship Diagram (10)

Total Marks = 40

Percentage = $40/200 \times 100 = 20\%$

Project Work (40 %)

- Front-end and GUI (10)
- Coding (20)
- Design patterns and best practices (10)
- S.O.L.I.D principles (10)
- Unit Testing (10)
- Integration Testing (10)
- End-to-end testing (10)

Total Marks = 80

Percentage = $80/200 \times 100 = 40\%$

Results and Presentation (20%)

- Documentation 2 (10)
- Documentation 3 (10)
- Documentation 4 (10)
- Project Exhibition (10)

Total Marks = 40

Percentage = $40/200 \times 100 = 20\%$

7. FYP Roadmap

FYP master plan can be seen in Figure 2, explaining the step by step process starting from 6th semester to 8th semester.

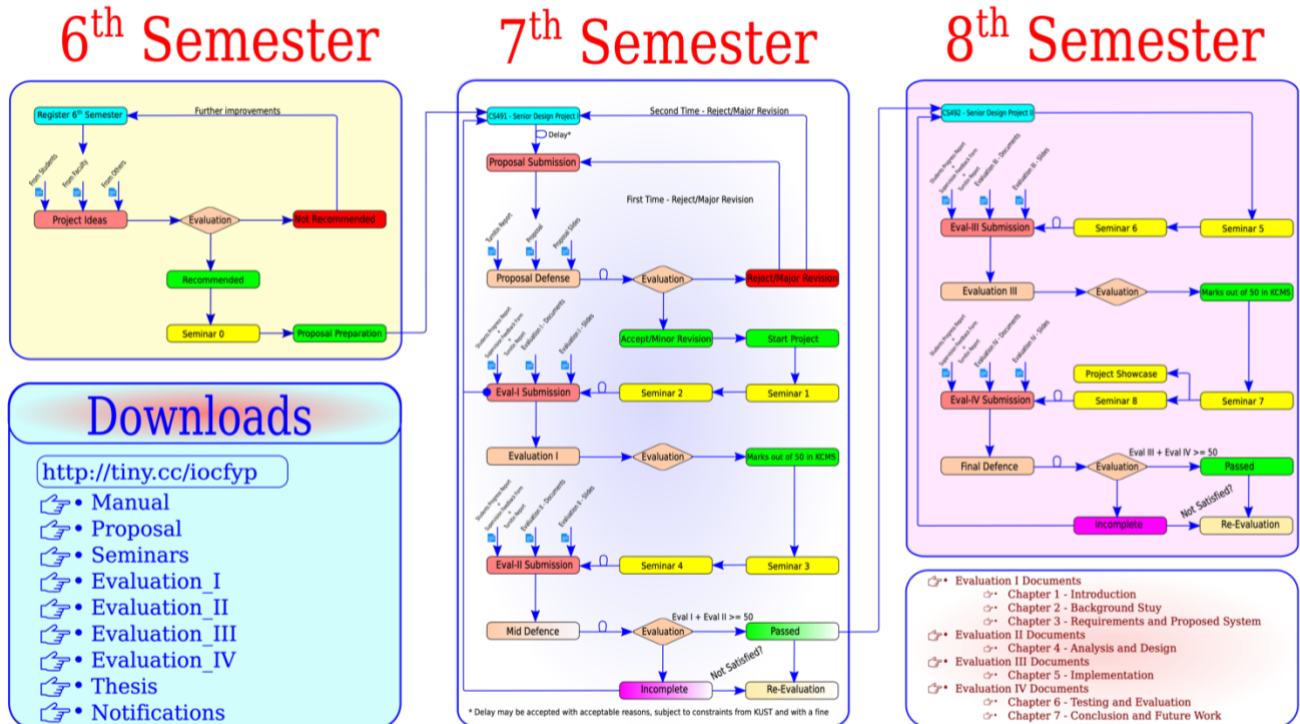


Figure 2: FYP road map

8. Seminars

The role of FYP panel is not only to evaluate students, but provide guidance related to their projects from time to time. During FYP session eight seminars related to different topics of projects will be arranged. Four seminars will be arranged in seventh semester and four seminars in the eighth semester. These seminars will be arranged in the third, sixth, ninth and thirteen weeks. Details of the seminars can be found in table below.

S.No.	Seminar Topic	Proposed Presenter	Date
1.	FYP Seminar	FYP Panel	4 th week 6 th semester
2.	How to write thesis?	Dr. Amjad Mehmood	3 rd week 7 th semester
3.	Software Development lifecycle and models	Mr. Rooman	6 th week 7 th semester
4.	Software Design and Analysis	Mr. Qadeem Khan	9 th week 7 th semester
5.	Front end and GUI	Mr. Qadeem Khan	12 th week 7 th semester
6.	Design patterns and best practices	Dr. Jawad Ashraf	3 rd week 8 th semester
7.	S.O.L.I.D. Principles	Dr. Jawad Ashraf	6 th week 8 th semester
8.	Unit Testing, Integration Testing and End-to-end Testing	Mr. Rooman	9 th week 8 th semester
9.	How the skills you learnt in FYP will help you in future?	Dr. Abdul Shahid	12 th week 8 th semester



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9. Evaluation I

Objectives

In this phase of the project, students will work on understanding the system to be developed. They will understand the problems to automate and requirements of the system to be developed and the objectives that they want to achieve in the proposed system. They will develop the design of the proposed system.

Submission

In the eighth week of the seventh semester, students will submit ring binding of Chapters 1 – 3 along with [Supervision Feedback Form \(By Students\)](#) (c.f. [appendixes](#)) and [Students Progress Report \(By Supervisor\)](#) (c.f. [appendixes](#)). Students will also submit Turnitin report signed by supervisor.

Although presentation is optional in this phase, students must submit slides explaining the topics required for [Evaluation I Form](#). Evaluation by all panel members are averaged in the form given in [Evaluation Decision Form](#). A template document and presentation slides are prepared for students to follow in directory [Evaluation I](#).

Evaluation

Panel will evaluate the submitted documents in the ninth week of seventh semester and assign marks out of 50 in KCMS for CS491. Students presentation for Evaluation I is optional. If Panel deems fit, students presentations will be arranged.

Topics to Study

The followings topics will help you to prepare yourself for Evaluation I.

1. How to initiate a project?

Resources:

- http://final-year-projects.com/Get_start.htm
- <https://www.theengineeringprojects.com/2018/12/final-year-project-the-ultimate-guide-for-beginners.html>

2. Software Development Life Cycle

Resources:

- <https://www.guru99.com/software-development-life-cycle-tutorial.html>
- <https://stackify.com/what-is-sdlc/>
- <https://www.softwaretestinghelp.com/software-development-life-cycle-sdlc/>
- <https://raygun.com/blog/software-development-life-cycle/>
- <https://searchsoftwarequality.techtarget.com/definition/systems-development-life-cycle>

3. Software Development Models

Resource:

- <https://melsatar.blog/2012/03/15/software-development-life-cycle-models-and-methodologies/>
- <https://medium.com/globaluxsoft/5-popular-software-development-models-with-their-pros-and-cons-12a486b569dc>
- <https://medium.com/omarelgabrys-blog/software-engineering-software-process-and-software-process-models-part-2-4a9d06213fdc>
- <http://mrbool.com/what-are-different-software-development-models/30120>
- <https://existek.com/blog/sdlc-models/>

4. Mock-up

Resources:

- <https://99designs.com/blog/tips/dos-and-donts-of-design-mockups/>
- <http://psd.fanextra.com/articles/a-comprehensive-guide-to-mockups-in-web-design/>
- <https://designmodo.com/mockups/>
- <https://www.justinmind.com/blog/guide-website-mobile-app-mockup/>
- <https://www.slideshare.net/marcelogracioli/the-guide-to-mockups>
- <https://brainhub.eu/blog/difference-between-wireframe-mockup-prototype/>
- <https://www.invisionapp.com> -> Better use this tool. With KUST email, u get free access



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5. Data Flow Diagram

Resources:

- <https://www.lucidchart.com/pages/data-flow-diagram>
- <https://www.geeksforgeeks.org/levels-in-data-flow-diagrams-dfd/>
- <https://www.sciencedirect.com/topics/computer-science/data-flow-diagram>
- <https://medium.com/@warren2lynch/data-flow-diagram-comprehensive-guide-with-examples-d9858387f25e>

6. Class Diagram

Resources:

- <https://www.lucidchart.com/pages/uml-class-diagram>
- <https://www.guru99.com/uml-class-diagram.html>
- <http://agilemodeling.com/artifacts/classDiagram.htm>

7. How to write thesis/document?

Resources:

- <https://batchloaf.wordpress.com/2012/05/02/some-tips-on-writing-a-final-year-project-report/>
- <http://mjiit.utm.my/wp-content/uploads/2018/09/FYP-1-How-to-Write-Thesis.pdf>
- <https://pats.cs.cf.ac.uk/wiki/lib/exe/fetch.php?media=project-report.pdf>
- https://www.researchgate.net/publication/301198221_Student's_Guide_For_Final_Year_Project_Thesis_BSc_MSc_MA_and_MBA

Note: Even though DFD is not part of Object Oriented Design, we think that it is important that students are able to create DFDs at a more refined level. This process will help them to better understand the system.

10. Mid Defence/Evaluation II

Objectives

In this phase of the project student will complete the design of the proposed system. They will also work on the front end implementation of the system.

Submission

In the Sixteenth week of the seventh semester students will submit ring binding of Chapter 4 along with [Supervision Feedback Form \(By Students\)](#) (c.f. appendixes) and [Students Progress Report \(By Supervisor\)](#) (c.f. appendixes). Students will also submit Turnitin report signed by supervisor. A template documentation and presentation slides are prepared for students to follow in directory [Evaluation II](#).

Evaluation

Panel will schedule mid defence presentation soon after final exams are finished. Students are evaluated based on the parameters mentioned in [Evaluation II Form](#). Evaluation by all panel members are averaged in the form given in [Evaluation Decision Form](#). At this stage the remaining 50 marks are entered in KCMS. After this evaluation, students have 100 marks of CS491 submitted in KCMS.

Topics to Study

The following topics will help you to prepare yourself for Evaluation II.

1. Use-case Diagram

Resources:

- <https://www.lucidchart.com/pages/uml-use-case-diagram>
- <https://www.uml-diagrams.org/use-case-diagrams.html>
- <http://www.agilemodeling.com/artifacts/useCaseDiagram.htm>
- <https://createely.com/blog/diagrams/use-case-diagram-tutorial/>

2. Activity Diagram

Resources:

- <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-activity-diagram/>
- <https://www.smartdraw.com/activity-diagram/>



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- <https://www.geeksforgeeks.org/unified-modeling-language-uml-activity-diagrams/>
- <https://www.uml-diagrams.org/activity-diagrams.html>

3. Sequence Diagram

Resources:

- <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-sequence-diagram/>
- <https://www.smartdraw.com/sequence-diagram/>
- <https://www.geeksforgeeks.org/unified-modeling-language-uml-sequence-diagrams/>
- <http://www.agilemodeling.com/artifacts/sequenceDiagram.htm>

4. Entity Relationship Diagrams

Resources:

- <https://www.smartdraw.com/entity-relationship-diagram/>
- <https://www.visual-paradigm.com/guide/data-modeling/what-is-entity-relationship-diagram/>
- <https://www.guru99.com/er-diagram-tutorial-dbms.html>
- <https://beginnersbook.com/2015/04/e-r-model-in-dbms/>

5. Front End and GUI

Resources:

- <https://vintage.agency/blog/front-end-developer-vs-ui-developer-who-is-who/>
- <https://www.guru99.com/frontend-testing.html>
- <https://frontendmasters.com/books/front-end-handbook/2019/>
- <https://frontendmasters.com/books/front-end-handbook/2018/what-is-a-FD.html>
- <https://www.computerhope.com/jargon/g/gui.htm>
- <https://www.itpro.co.uk/operating-systems/30248/what-is-a-graphical-user-interface>
- <https://www.amazon.com/dp/1568843224/>
- <https://www.thoughtco.com/gui-2034108>
- https://www.ntu.edu.sg/home/ehchua/programming/java/J4a_GUI.html
- <https://study.com/academy/lesson/developing-a-graphical-user-interface-gui.html>

Note: ERD may not be possible when a relational database is not used. In that case, the non-relational model will be evaluated.

11. Evaluation III

Objectives

In this phase of the project, students will work on the implementation of the system to be developed. They will develop a complete system, that will be able to achieve the objectives given in proposal.

Submission

In the mid of the eighth semester students will submit ring binding of Chapter 5 along with [Supervision Feedback Form \(By Students\)](#) (c.f. appendixes) and [Students Progress Report \(By Supervisor\)](#) (c.f. appendixes). Students will also submit Turnitin report signed by supervisor.

Although presentation is optional in this phase, students must submit slides explaining the topics required for [Evaluation III Form](#). Evaluation by all panel members are averaged in the form given in [Evaluation Decision Form](#). A template document and presentation slides are prepared for students to follow in directory [Evaluation III](#).

Evaluation

Panel will evaluate the submitted documents in the ninth week of eighth semester and assign marks out of 50 in KCMS for course CS492. Students presentation for Evaluation III is optional. If Panel deems fit, students presentations will be arranged.

Although presentation is optional in this phase, students must submit slides explaining the topics covered in the [Evaluation III Form](#). Evaluation by all panel members are averaged in the form given in [Evaluation Decision Form](#). A template presentation slides is prepared for students to follow ([Evaluation III/Evaluation III.pptx](#)).



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Topics to Study

The followings topics will help you to prepare yourself for Evaluation III.

1. Design patterns

Like MVC, MVP, MVVM, BloC etc.

Resources:

- <https://proandroiddev.com/mvc-mvp-mvvm-clean-viper-redux-mvi-prnsaaspfrucc-building-abstractions-for-the-sake-of-building-18459ab89386>
- <https://medium.com/flutterpub/architecting-your-flutter-project-bd04e144a8f1>
- <https://medium.com/flutter-community/flutter-bloc-pattern-for-dummies-like-me-c22d40f05a56>
- https://sourcemaking.com/design_patterns
- <https://refactoring.guru/design-patterns>
- <https://www.geeksforgeeks.org/software-design-patterns/>

2. Coding techniques and rules

Separation of concerns, single responsibility, DRY, KISS, YAGNI, etc.

Resources:

- <https://deviq.com/separation-of-concerns/>
- <https://medium.com/machine-words/separation-of-concerns-1d735b703a60>
- <https://medium.com/datadriveninvestor/programming-fundamentals-part-5-separation-of-concerns-software-architecture-f04a900a7c50>
- <https://www.toptal.com/software/single-responsibility-principle>
- <https://hackernoon.com/you-dont-understand-the-single-responsibility-principle-abfdd005b137>
- <https://medium.com/@severinperez/writing-flexible-code-with-the-single-responsibility-principle-b71c4f3f883f>
- <https://dzone.com/articles/is-your-code-dry-or-wet>
- <https://dzone.com/articles/software-design-principles-dry-and-kiss>
- <https://dev.to/wuz/stop-trying-to-be-so-dry-instead-write-everything-twice-wet-5g33>
- <https://code.tutsplus.com/tutorials/3-key-software-principles-you-must-understand--net-25161>
- <https://thevaluable.dev/dry-principle-explained/>
- <https://metova.com/dry-programming-practices/>
- <https://martinfowler.com/bliki/Yagni.html>
- <https://deviq.com/yagni/>
- <https://www.itexico.com/blog/software-development-kiss-yagni-dry-3-principles-to-simplify-your-life>
- <https://medium.com/@derodu/design-patterns-kiss-dry-tda-yagni-soc-828c112b89ee>

3. S.O.L.I.D Principle

Resources:

- <https://siderite.blogspot.com/2017/02/solid-principles-plus-dry-yagni-kiss-final.html>
- <https://scotch.io/bar-talk/s-o-l-i-d-the-first-five-principles-of-object-oriented-design>
- <https://hackernoon.com/solid-principles-simple-and-easy-explanation-f57d86c47a7f>
- <https://hackernoon.com/solid-principles-made-easy-67b1246bcd>
- <https://williamdurand.fr/2013/07/30/from-stupid-to-solid-code/>
- <https://dzone.com/articles/solid-grasp-and-other-basic-principles-of-object-o>

12. Final Defence/Evaluation IV

Objectives

In this phase of the project student will implement and test the proposed system. They will also work on the implementation, testing and integration of the system.

Project Exhibition

In the thirteenth month of eighth semester a project exhibition will be organized at Institute of Computing. Students are advised to make banner (pane flex 2.5 x 6 feet) of your project and bring on the notified day. All details of the exhibition will be shared with students. 10 out of 50 marks are assigned for this exhibition in the final evaluation.



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Submission

In the Sixteenth week of the eighth semester, students will submit ring binding of Chapters 6 and 7 along with [Supervision Feedback Form \(By Students\)](#) (c.f. [appendixes](#)) and [Students Progress Report \(By Supervisor\)](#) (c.f. [appendixes](#)). Students will also submit Turnitin report signed by supervisor. A template documentation and presentation slides are prepared for students to follow in directory [Evaluation IV](#).

Evaluation

Panel will schedule final defence presentation soon after final exams are finished. Students are evaluated based on the parameters mentioned in [Evaluation IV Form](#). Evaluation by all panel members are averaged in the form given in [Evaluation Decision Form](#). At this stage the remaining 50 marks are entered in KCMS. After this evaluation, students have 100 marks of CS492 submitted in KCMS.

Topics to Study

The followings topics will help you to prepare yourself for Evaluation IV.

1. Automated testing

Resources:

- <https://searchsoftwarequality.techtarget.com/definition/automated-software-testing>
- <https://www.guru99.com/automation-testing.html>
- <https://medium.com/@briananderson2209/best-automation-testing-tools-for-2018-top-10-reviews-8a4a19f664d2>
- <http://www.softwaretestingclass.com/what-is-automation-testing/>

2. Unit testing

Resources:

- <https://programmingwithmosh.com/net/unit-testing/>
- <https://www.guru99.com/unit-testing-guide.html>
- <http://softwaretestingfundamentals.com/unit-testing/>

3. Integration Testing

Resources:

- <http://softwaretestingfundamentals.com/integration-testing/>
- <https://www.guru99.com/integration-testing.html>
- <https://www.softwaretestinghelp.com/what-is-integration-testing/>

4. End-to-end testing

Resources:

- <https://www.guru99.com/end-to-end-testing.html>
- <https://www.softwaretestinghelp.com/what-is-end-to-end-testing/>
- <https://smartbear.com/learn/automated-testing/how-to-perform-end-to-end-testing/>
- <https://dzone.com/articles/what-is-end-to-end-testing-1>
- <https://dzone.com/articles/all-you-need-to-know-about-end-to-end-testing>

13. Project Showcase

In thirteenth week of the eighth semester, a project showcase will be organized. Where students will display their projects and people from within the institute as well as outside will be invited to visit the projects and ask questions. The idea is to promote these projects at university level and at the same time attract industry to potentially establish industry linkages.

We have assigned 10 marks in Evaluation IV for project exhibition. We will ask external evaluators (4-5) to evaluate projects i.e. none of the faculty member of IoC will be involved in evaluation. The idea is to check if students are able to provide information about their projects to nontechnical people. The exhibition will be arranged in the corridors or labs of Institute of Computing building. All the posters are displayed on the wall in corridor or with the help of standees.



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VC is invited as chief guest. We will give certificates and cash prizes to 1st, 2nd and 3rd position holders. 1st prize is 6000 Rs, 2nd prize is 4000 Rs and 3rd prize is 2000 Rs. This is a one day activity.

14. Thesis Template

Chapter 1 – Introduction

Chapter 2 – Background/Literature

- This chapter must also include SDLC and Software Development Model of your project

Chapter 3 – Requirements and proposed system

Chapter 4 – Design and analysis

Chapter 5 – Implementation

Chapter 6 – Testing, Integration and Evaluation

Chapter 7 – Conclusion and Future Work

A separate document of thesis is provided to students. (File name is [Thesis/Thesis_Template.docx](#))

When get passing marks in CS491 and CS492, they are asked to prepare **THREE** hard bounded thesis copies and submitted to the office of Director IoC.

15. Appendixes

All forms/documents/template used during FYP are attached here.



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Project Idea Submission Form

Date: _____

Students Name

(All members in group)

Supervisor

Plan A

☐

Plan B

☐

Project Title

Project Description

<type here. Maximum 250 words>

Tool and Technologies to be used:

Comments by panel member(s)

Decision by Panel

Recommended

☐

Not Recommended

☐

Signatures:



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Proposal Evaluation Form

Date: _____

Student Name(s)				
Title				
Panel Member Name				
S.No.	Parameters evaluated	Below Average (Range: 0-4.9)	Average (Range: 5-6.9)	Above Average (Range: 7-10)
1.	Problem Identification - Is the problem well identified? - Have students put effort to make it clear?			
2.	Problem Understanding - Is the problem well understood? - Have students made effort in understanding?			
3.	Background Study - Have students explored existing methods/technologies? - Have students compared the idea with existing technologies?			
4.	Design and Analysis Effort - Does the idea require enough design and analysis? - Is the effort justified for the number of students?			
5.	Implementation Effort - Does the idea require enough implementation? - Is the work justified for 3-6 months per person?			
6.	Evaluation/Testing/Comparison - Does the idea require enough evaluation/testing? - Is the need and effort of testing justified?			
7.	Three-six months effort per person - Does the idea require enough implementation effort? - Is the size of the project justified over the number of students?			
8.	Latest Technology - Are students exploring new and latest technologies? - Are students familiar with recent advancement in the field?			
9.	Proposal Documentation - Is the proposal well documented and in a good flow? - Are there typos, grammatical errors and plagiarised contents? - Is Turnitin report signed by supervisor attached?			
10.	Presentation Skills - Are students communicating in clear and organized style? - Are students convincing and enthusiastic?			
Total				

Comments

Signature by Panel Member: _____



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Proposal Decision Form

Date: _____

Student Name(s)		
Title		
First time Defence ☐	Second time Defence ☐	

Calculation

Marks by Panel Member 1	Marks by Panel Member 2	Marks by Panel Member 3	Marks by Panel Member 4	Marks by Panel Member 5	Average Marks

Decision by FYP Panel

Approved ☐	Minor Revisions ☐	Major Revisions ☐	Not Approved ☐
-----------------------------------	--	--	---------------------------------------

Notes:

- At least three members will evaluate the proposal.
- For approved as it is, at least 70% marks on average are obtained.
- For minor revisions, at least 60% marks on average are obtained.
- For major revisions, at least 50% marks are obtained.
- For major revisions and not approved cases, a second defence will be arranged.
- Students getting major revisions and not approved for the second time, will re-register the semester.

Signature by FYP Coordinator



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Student Progress Report (By Supervisor)

Date: _____

S1	
S2	
S3	
Title	

Please fill the following table.

S.NO.	Question	Your response
1.	Do students meet you bi-weekly?	
2.	Are you satisfied with their work so far?	
3.	Are they following your guidelines	
4.	Are they putting effort in learning?	
5.	Have you checked their report?	

Please provide us the meeting details

S.NO.	Date	No. of Students present	What Topics you discussed?
1.			
2.			
3.			
4.			
5.			

How do you rate students on the scale of 0-10?

S1	
S2	
S3	

Please provide us detailed information about students' progress

Signature and name of Supervisor: _____

Note: This form must be submitted with every Evaluation.



Date: _____

Please fill the following table.

Please provide us the detail of bi-weekly activities

How do you rate your supervisor on the scale of 0-10?

Please provide us detailed information about your activities and your interaction with supervisor

Note: This form must be submitted with every Evaluation.



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Evaluation I Form

Date: _____

S1	
S2	
S3	
Title	

Performance	Poor	Fair	Good	Excellent
Marks Range	(0 – 4.9)	(5 – 6.9)	(7 – 8.9)	(9– 10)

S.No.	Evaluation Parameters	S1	S2	S3
1.	SDLC? - Do students understand the whole process of SDLC? - Have they developed the SDLC of their own project?			
2.	Software Development Model? - Are students familiar with different software development models? - Have they identified the model they are following in their project?			
3.	Mock-ups (Full system) - Have they prepared mock-ups of full system? - Is a standard procedure followed in the design of mock-ups?			
4.	Data Flow Diagrams - Are there DFDs from 0 to more detailed levels (level 2)? - Is it easy to understand the behaviour of the whole system? Class Diagram (single diagram) - Are all classes shown in the diagrams? - Is it easy to understand the structure of the whole system?			
5.	- Students' Progress Report (By Supervisor) - Supervision Feedback Form (By Students) - Turnitin report - Documentation 1 - Chapter 1 – Introduction - Chapter 2 – Background Study - Chapter 3 – Requirements and proposed system			
Total (50 Marks)				

Comments

Signature and name of Panel Member: _____



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Evaluation II Form

Date: _____

S1	
S2	
S3	
Title	

Performance	Poor	Fair	Good	Excellent
Marks Range	(0 – 4.9)	(5 – 6.9)	(7 – 8.9)	(9– 10)

S.No.	Evaluation Parameters	S1	S2	S3
1.	Use-case Diagram (single diagram) - Is the functionality of the system explained? - Is it easy to understand the functionality of the whole system from this diagram? Activity diagrams (multiple diagrams) - Is the flow from activity to activity explained? - Are workflow/business process and internal operation explained?			
2.	Sequence diagrams (multiple diagrams) - Are the interaction amongst classes explained? - Is it easy to understand the sequence of information from these diagrams?			
3.	Entity Relationship Diagram (Single diagram) - Are entities, attributes and their relationships clearly presented? - Is it easy to understand the structure of the whole database?			
4.	Front-end and GUI - Have they implemented front-end? - Is the coding implementing what is explained in different diagrams?			
5.	- Students' Progress Report (By Supervisor) - Supervision Feedback Form (By Students) - Turnitin report - Documentation 2 - Chapter 4 – Analysis and Design			
Total (50 Marks)				

Comments

Signature and name of Panel Member: _____

Note: In case non-relational database is used, then the design of related model is evaluated instead of ERD



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Evaluation III Form

Date: _____

S1	
S2	
S3	
Title	

Performance	Poor	Fair	Good	Excellent
Marks Range	(0 – 4.9)	(5 – 6.9)	(7 – 8.9)	(9– 10)

S.No.	Evaluation Parameters	S1	S2	S3
1.	Written Code - Is the code compact, clean, readable and commented? - Is the coding justified for FYP?			
2.	Update in Code - Do they understand the code they have written? - Are they able to change something in the code?			
3.	Design patterns and best practices - Are design patterns like MVC, MVP, MVVM, Redux etc. followed? - Are rules like DRY, KISS, YAGNI and separation of concerns followed?			
4.	S.O.L.I.D. Principles - Are S.O.L.I.D. principles followed?			
5.	- Students' Progress Report (By Supervisor) - Supervision Feedback Form (By Students) - Turnitin report - Documentation 3 - Chapter 5 – Implementation			
Total (50 Marks)				

Comments

Signature and name of Panel Member: _____



Kohat University of Science and Technology

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Evaluation IV Form

Date: _____

S1	
S2	
S3	
Title	

Performance	Poor	Fair	Good	Excellent
Marks Range	(0 – 4.9)	(5 – 6.9)	(7 – 8.9)	(9– 10)

S.No.	Evaluation Parameters	S1	S2	S3
1.	Unit Testing - Are the tests well-written - Are the tests automated			
2.	Integration Testing - Are the tests well-written - Are the tests automated			
3.	End-to-end testing - Is end-to-end testing performed? - Are tests automated?			
4.	Project Showcase 2020 - Did students participate in project showcase? - What was the score by judges?			
5.	- Students' Progress Report (By Supervisor) - Supervision Feedback Form (By Students) - Turnitin report - Documentation 4 - Chapter 6 – Testing and Evaluation - Chapter 7 – Conclusion and Future Work			
Total (50 Marks)				

Comments

Signature and name of Panel Member: _____



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Evaluation Decision Form

Date: _____

S1	
S2	
S3	
Title	

Calculation

S.NO.	Marks by Panel Member 1	Marks by Panel Member 2	Marks by Panel Member 3	Marks by Panel Member 4	Marks by Panel Member 5	Average Marks
S1						
S2						
S3						

Notes:

- At least three members must be present during the evaluation.
- Evaluation by less than three members is not acceptable.

Signature by FYP Coordinator



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Late Submission/Defence Form

Date: _____

Student Name	
Project Title	

Reason of Delay

--

Please tick on the following

☐ Medical ☐ Death in the family ☐ Accident ☐ Others

Signature of Student _____

Approved ☐ Not Approved ☐

Signature of Director IoC: _____

Fine Amount: _____

Signature of FYP Coordinator: _____



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Group Change Form

Date: _____

Student Name	
Project Title	

Current Group Members

S1	
S2	

New Group Members

S1	
S2	

Reason of Change

--

Signature of Student _____

Decision of the Panel

Approved ☐ Not Approved ☐

Signature of FYP Panel Members: _____

Signature of FYP Coordinator: _____



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Supervisor Change Form

Date: _____

Students Name	
Project Title	

Current Supervisor Name

Have you discussed this with current supervisor?	

New Supervisor Name

Have you discussed with new supervisor?	

Reason of Change

--

Signature of Students _____

Approved ☐ Not Approved ☐

Signature of Director IoC: _____

Signature of FYP Coordinator: _____



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Panel Member Objection Form

Date: _____

Students Name	
Project Title	

Name of Panel Members Students have objection

Panel Member 1	
Panel Member 2	

Reason of Objection

--

Signature of Students _____

Approved ☐ Not Approved ☐

Signature of Director IoC: _____

Signature of FYP Coordinator: _____



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Re-Evaluation Form

Date: _____

Students Name	
Project Title	

Marks Obtained in CS491/CS492

Students Name	Evaluation I	Evaluation II	Evaluation III	Evaluation IV	Total

Reasons of Re-Evaluation

--

Signature of Students _____

Fee of 500 Rs per Student submitted with FYP Coordinator? Yes/No

Signature of FYP Coordinator: _____

Forwarded for Re-evaluation: ☐

Rejected: ☐

Signature of Director IoC: _____



Kohat University of Science and Technology

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Re-Evaluation Decision Form

Date: _____

Students Name	
Project Title	

Comments

--

Grade Change Approved: ☐

Grade Change not Approved: ☐

Recommended Grade (if approved):

--

Signature of Re-Evaluation Committee: _____



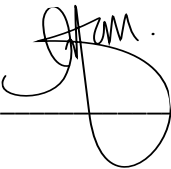
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16. Manual Approval

The document has been thoroughly studied and unanimously approved for the FYP activities and evaluation at Institute of Computing, KUST. This manual will be used as an official document for FYP starting from Fall Semester 2019.

1. **Dr. Muhammad Irfan Uddin** – Coordinator FYP Panel

Signature: 

Date: September 23, 2019

2. **Dr. Shafaq Mussadiq** – Member FYP Panel

Signature: _____

Date: September 23, 2019

3. **Dr. Muneer Umar** – Member FYP Panel

Signature: _____

Date: September 23, 2019

4. **Mr. Muhammad Roman** – Member FYP Panel

Signature: _____

Date: September 23, 2019

5. **Mr. Muhammad Ali Zeb** – Member FYP Panel

Signature: _____

Date: September 23, 2019

6. **Dr. Shafiullah** – Director Institute of Computing

Signature: _____

Date: September 23, 2019